

The Gate and the First Flow: Program ω , Stage C's Opening Move

Harald Rønning

Revision 1.1 — still open, discriminator 2 and the interpretation session pending. Companion to the frozen ω -plan (Stage B closed on the two-sign question; this note covers the Gaussianity gate, the arithmetic fork it surfaced, the first flow integration on two anchors — now two-implemented — and the frozen-sector identification the flow itself surfaced). Every claim below is two-implemented unless explicitly marked otherwise; the error ledger is part of the record because the catches, across this note's two revisions, are the substance of what was learned.

1. Charter

Stage B closed with $m(u)$'s asymptotic signs undetermined-without-dynamics — the missing datum handed forward to Stage C. This note covers Stage C's own opening: constructing the Gaussianity gate the frozen plan's P3 amendment specifies, the extended arithmetic fork that construction surfaced, and the first flow integration once the fork closed. It ends without a verdict, by design — the two anchors integrated here produced genuinely different, only partially interpreted behavior, and that openness is the honest state of the record, not a placeholder for a conclusion not yet written.

2. The Gaussianity gate: construction and its two-implementation

[FACT, two-implemented] The candidate dissipative dynamics is built from linear jump operators on the anchor's own real-linear operator L_A , expressed in a mode basis built from $J_0 = -L_{e_4} - \mathbb{S}$'s own invariant complex structure, fixed by the mediator direction, distinct from any wall-sector or gauge-module object also carrying the name " J_0 " elsewhere in the corpus. (This distinction is stated explicitly here because blurring it was the exact species that caused this arc's own stale-coordinate bug, §3 below; a note about that bug should not commit it.) In quadratures $r = (x_1, p_1, \dots)$, each jump $L_j = c_j^T r$, with $M = \sum_j c_j c_j^\dagger$, the covariance obeys the closed-form Lyapunov equation

$$\dot{\sigma} = A\sigma + \sigma A^T + D, \quad A = -\Omega \operatorname{Im}(M), \quad D = \Omega \operatorname{Re}(M) \Omega^T,$$

σ centered and symmetrized, Ω the standard symplectic form. This formula was independently derived and cross-verified against direct Fock-space simulation (finite truncation, exact master-equation dissipator), converging to machine precision as truncation grows — the formula itself was never in dispute once both sides tested it against ground truth rather than each other.

3. The arithmetic fork, and what it actually was

[LEDGER, the arc's own record] What followed was not a physics disagreement but a five-layer convention stack, each layer catching a genuine error before the next could hide behind it.

Layer 1 — stale coordinates. The anchor L_A built fresh in this session's own basis was combined with a saved J_0 file from an earlier part of this long session, built on a silently different basis realization. Both objects passed every intrinsic check individually ($J_0^2 = -\text{Id}$ exact; L_A 's tunnel spectrum exact) — alignment between them was never tested until a full rebuild on consistent inputs was forced. Root cause named precisely: **saved-object coordinate provenance** — a file can carry validity without carrying the coordinate frame it was validated in, and two individually-correct objects from different moments of one session are not automatically combinable. Resolution: full rebuild from one `cd16` call, no saved files. Confirmed split: $\|C\|^2 = 4$, $\|D\|^2 = 12$, matching the independently-derived basis-free invariant ($\|L_c\|^2/2$, $\|L_a\|^2/2$ for the J_0 -complex-linear/antilinear parts of L_A) exactly.

Layer 2 — the definitional fork. Two constructions for “the anchor's dynamics under the mirror orientation” were each built and verified exact: (a) the same quadrature machine fed the mirror's conjugated inputs, giving $A_2 = A_1$ (orientation-blind) and $[P, A_1] = 0$; (b) the literal adjoint jump-set $\{L_j^\dagger\}$, giving $A_2 = -A_1$ (arrow-duality, $\text{loss} \leftrightarrow \text{gain}$). Both are exact theorems about two different, well-defined objects — not competing answers to one question. **[THM, resolved as a duality, not a choice]** The two compose: orientation-flip followed by the construction equals time-reversal of the construction. Neither reading is privileged; a future document naming “the mirror candidate” states locally which half it means.

Layer 3 — a mislabeled scalar. A proposed “kernel diffusion = $N(A)$ ” identity was traced to its source and found to be a Frobenius norm of a 4×4 block, reported as a per-direction rate. Rerun with the value computed two independent ways (via the assembled D matrix, and directly as $\sum_j |c_j \cdot k|^2$) gives $k^T D k = 1.0$ per kernel direction, both routes agreeing to machine precision. Bonus structural fact, confirmed: $\text{Re}(M)$ annihilates the kernel directly — the jump vectors are exactly orthogonal to it — so the diffusion the kernel experiences arrives entirely through the Ω -dressing, from its symplectic partner directions, not from any direct coupling.

Species logged, this arc, both sides: *saved-object coordinate provenance* (mine); *output-attribution blur*, *rationalized disconfirmation*, *theorem-scope inflation*, *quantity-type mislabeling* (F's, each independently self-diagnosed and ledgered in turn). The countermeasure common to all: a cheap, mechanical check (print the actual matrices consumed, not re-derived ones) run *before* trusting an internally-consistent-looking result, not after a contradiction forces

it.

4. Six results, closed

Confirmed two-implemented, on fully consistent inputs, no open threads:

1. **Split:** $\|C\|^2 = 4$, $\|D\|^2 = 12$, $\text{tr}(A) = 8$.
2. **Drift spectrum:** $\{0^8, +1^8\}$ for the classic anchor $e_1 + e_{10}$.
3. **D-equality:** $D_2 = D_1$ under either reading of the orientation flip.
4. **Kernel drift-freeze:** exact to 10^{-16} , refined to $\text{Re}(M)$ -annihilation.
5. **The 84-diagonal survey:** exactly three classes, 12/60/12, trace $\in \{+16, +8, -8\}$.
6. **The strut cross-tabulation:** the mediator kite is uniquely class-pure (all 12 diagonals in the middle class); every other kite splits identically $2+8+2$ across the three classes, verified via the seam note's own assessor-product strut rule.

A seventh, discovered by the flow rather than the gate: the linear-growth branch's slope in the contracting anchor's own $\chi_{\text{kin}}(t)$ trajectory equals the gate's certified kernel diffusion constant exactly — 1 per direction per unit time — two independently-certified numbers, one from the gate's static analysis and one from the flow's dynamics, agreeing across the gate/flow boundary with no free parameter connecting them.

5. The first flow: two anchors, three discriminators, two-implemented

[FACT, two-implemented] Two anchors integrated from a shared quasi-free initial state ($\nu = 1.5$): the classic expanding-class anchor ($e_1 + e_{10}$, $\text{tr } A = +8$) and a contracting-class anchor ($e_4 + e_9$, $\text{tr } A = -8$, drift spectrum $\{-2^4, 0^{12}\}$ — a genuinely different structure from a simple sign-flip of the expanding case). An independent closed-form integration (a different ODE route — the exact matrix-exponential solution rather than numerical quadrature — and a different symplectic-spectrum implementation) reproduced every landmark digit-for-digit: the expanding branches $\{1.534, 6.078\} \rightarrow \{2.373, 1139.067\}$; the contracting three-branch structure with multiplicities $2/2/4$ — the vacuum-landing at exactly 0.5, the branch frozen at exactly 1.5 throughout, and the linear branch as $\nu = 1.5 + t$ exactly; the family-residual growth rates, consistent up to metric convention.

Discriminator 1 (deepens vs. wanders), stated as result. The expanding anchor's χ_{kin} **deepens:** both branches grow monotonically, every successive state comparable and strictly deeper in Becoming's own partial order — genuine, coordinated, spectrum-valued depth-motion, off-spine, without ever reaching stationarity. The contracting anchor **wanders:** its three branches move incomparably — one decaying, one static, one growing — simultaneously

in all three senses, which is wandering by the same criterion applied honestly. The anchor named for stability (by its drift matrix) is the one that wanders; the anchor that deepens cleanly never rests. Given the corpus's own direction convention (the vacuum end as the unattained past), the expanding anchor's behavior is deepening without completion — the no-completion doctrine, asserted as state-theory since Foundation's own boundary facts, realized here by a dynamics nobody tuned to produce it.

Discriminator 3 (family-tracking). Neither anchor stays on the quasi-free line; both depart it immediately upon integration, at rates differing by roughly two orders of magnitude — the expanding anchor's departure explodes, the contracting anchor's is comparatively gentle.

6. The frozen sector: identified, mechanism included

[FACT, two-implemented] The contracting anchor's flow carries a genuinely silent sector — four directions with both zero drift and zero diffusion, frozen at their initial value throughout the integration. A neutral scan against every natural candidate subspace (no favored hypothesis) identifies these exactly: **the frozen directions coincide precisely with L_A 's own kernel** — the seam note's tunnel — with maximum overlap (2.0000 against a 4-dimensional space) and exact zero against every other candidate tested (the scalar span, the anchor plane, the strut fiber). The split against the τ -halves is exactly equal, matching the tunnel lemma's own "perfectly τ -balanced" description precisely. The expanding anchor carries no such sector at all — every one of its zero-drift directions carries identical, nonzero diffusion; the frozen phenomenon is not generic to the construction.

[LEMMA, two-implemented] The mechanism. For $k \in \ker L_A$, $L_c k = -L_a k$ always (since $L_c + L_a = L_A$ vanishes on the kernel) — but the kernel is J_0 -invariant if and only if the stronger, joint condition holds: $L_c k = L_a k = 0$ *separately*. Verified across all 84 canonical diagonals: at the pure-antilinear class ($C^2 = 0$), this is automatic; at the genuinely mixed extreme class ($C^2 = 12, D^2 = 4$ — *not* pure, both parts nonzero), the joint annihilation is a real, nontrivial fact, confirmed exactly ($\|L_c \cdot \ker\| = \|L_a \cdot \ker\| = 0$ on all 12 members); at the generic middle class, it fails on all 60. **The seam note's own tunnel lemma — kernel commutes with neither τ nor L_{e_4} — is generic, not universal:** the exceptional set is exactly these 24 extreme-class diagonals, both classes, not one.

[FACT, two-implemented] Question (iv), answered honestly negative. The scalar directions (e_0, e_8) were tested directly against each anchor's own drift/diffusion structure, no hypothesis presumed: both carry nonzero drift and nonzero diffusion in both anchors. **The dissipation does not conserve the scalar channel** — it engages the measurement channel exactly as it engages any generic direction. The kernel avoids the scalars statically (the scalar-

avoiding kernel theorem stands, untouched); the dynamics does not.

7. Open items, carried forward

- Discriminator 2 (direction structure against the tunnel bands) on both trajectories — its interpretation is now pre-built, since the band decomposition and the branch anatomy are the same underlying skeleton, but the check itself has not yet run.
- Whether the contracting anchor’s three-way split (decay / frozen / linear growth) has a further structural explanation beyond the frozen sector now identified — the decay and linear branches remain uninterpreted.
- The interpretation session itself: with all four discriminators’ data assembled, whether either anchor’s behavior — or some structural feature short of either — constitutes “settling” in the corpus’s own sense has not yet been asked.
- Whether either anchor’s behavior should inform REG-2’s successor question (the anchor-supplier problem) remains entirely untouched by this note.
- Two downstream touches flagged, not yet made: the theorem inventory gains an entry for the kernel-invariance criterion (§6); the seam note’s tunnel lemma gains a one-line generic-not-universal scope refinement, at next opening.

Created in collaboration with Claude (Anthropic). Revision 1.1 — closed on the gate and the first flow; open on discriminator 2 and the interpretation session, as stated above. All conclusions and any errors remain the author’s responsibility.